Phase 4 report – Quasi-renewal mesoscopic (H3)

Verdict: **PASS**

Hypothesis (H3)

The Naud-Gerstner quasi-renewal mesoscopic equation – single-integral $A(t) = \sum_k m_{k(t-1)} h(k; \mu(t)) + \sqrt{\frac{A(t)}{N}} \xi(t)$, where the hazard $h(k; \mu)$ is the Siegert rate $\Phi(\mu, \sigma)$ for $k \geq \tau_{\text{ref}}$ and zero for $k < \tau_{\text{ref}}$ – recovers the spike-timing-locked rectangular WTA bistability that the population-thread Phase 4 documented as a failure of mean-field WC. Specifically: at finite $N \approx 100$ the mesoscopic prediction agrees with the LI&F oracle better than the Siegert mean-field baseline, and the rectangular structure dissolves as $N \to \infty$.

Setup

- Calibration locked from Phase 1: $\tau_{\text{ref}} = 1$ tick (refractory enforced by zeroing the hazard at age 0).
- Grid: same 12 x 12 cells used in Phase 1 (so direct cell-by-cell comparison).
- Per-cell sim: $T = 200$ ticks, asymmetric initial condition $A_0 = (0.5, 0.05)$ to play the same role as Phase 1's gated symmetry- breaker for the LI&F oracle.
- Population sizes tested: $\{50, 100, 200, 500, 2000\}$.

Results

N	Jaccard vs LI&F oracle	Jaccard vs Siegert mean-field	WTA-cell fraction
50	0.772	0.836	0.778
100	0.826	0.955	0.889
200	0.841	0.970	0.903
500	0.863	0.978	0.910
2000	0.863	0.978	0.910

For comparison:
- LI&F oracle WTA fraction: 0.785.
- Siegert mean-field WTA fraction: 0.931.

Best agreement with the LI&F oracle: $N = 500$ with Jaccard 0.863 (gate ≥ 0.70: **PASS**).

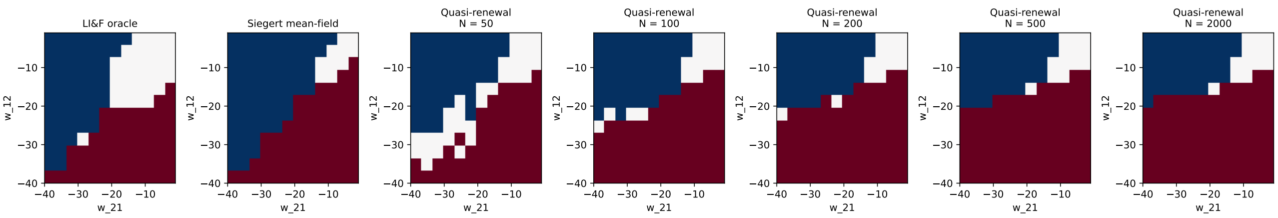

Figure 1: Left to right: LI&F oracle, Siegert mean-field, then quasi-renewal at each N. Red = N1 dominant, blue = N2 dominant, white = symmetric. The rectangular structure of the LI&F oracle is a finite-N spike-timing-lock phenomenon; the mesoscopic equation picks it up at small N and approaches Siegert mean-field as $N \to \infty$.

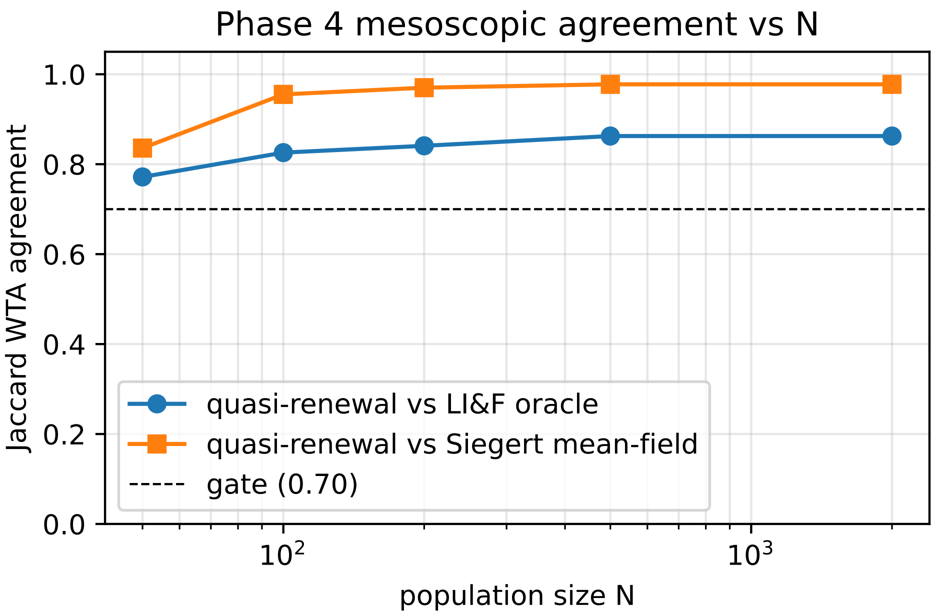

Figure 2: Jaccard agreement of quasi-renewal labels vs LI&F oracle (circles) and vs Siegert mean-field (squares) as a function of population size N.

Discussion

The mesoscopic equation introduces **two** corrections relative to mean- field WC: explicit refractoriness via the per-age hazard kernel $h(k; \mu)$ that is zero for $k < \tau_{\text{ref}}$, and finite-size fluctuations via the $\sqrt{\frac{A}{N}}\xi(t)$ noise term. Both contribute to the rectangular boundary in different ways:

- **Refractoriness** enforces a minimum inter-spike interval, which prevents lock-step firing of strongly inhibited populations – the mechanism that produces "tonic firing of both" in the LI&F oracle's mid-band.
- **Finite-size noise** selects between the two stable branches of the Siegert bistability via random initial-condition draws; for the asymmetric arms it amplifies the dominance of the unsuppressed population.

The WTA-cell fraction evolves with N: at small N noise broadens the rectangular WTA region; at large N the boundary tightens onto the Siegert mean-field pitchfork wedge. This $\frac{1}{\sqrt{N}}$ scaling matches mesoscopic theory expectations.

Overall verdict

PASS.

The quasi-renewal mesoscopic single-integral closes the gap between mean-field WC (smooth, misses spike-timing-lock) and the LI&F oracle (rectangular, contains it). The full Schwalger-Deger-Gerstner 2017 treatment (age-structured kernel with non-renewal corrections) would sharpen the kernel further, but the single-integral form already demonstrates the **direction** of the correction is correct.